SLUG PRESENTS

Sneak-Peak into Free & Open Source Software (FOSS) & GNU/Linux



A Comprehensive Guide for Engineering Students

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B block Seminar Hall

Sapthagiri NPS University

Agenda

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Definition, features & examples

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Definition, problems & limitations

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What is FOSS‘

Free & Open Source Software

Software where source code is freely available for anyone to view, modify, and distribute. "Free" means freedom, not price.

"Free as in free speech, not free as in cost"

- ✓ **Freedom to Use** - Run for any purpose
- ✓ **Freedom to Study** - Access source code
- ✓ **Freedom to Modify** - Change and improve
- ✓ **Freedom to Share** - Redistribute copies

Popular FOSS Examples



Linux Kernel
Operating system core



Mozilla Firefox
Web browser



VLC Media Player
Media player



LibreOffice
Office suite



Apache HTTP Server
Web server

Community-driven development with global collaboration

Features of FOSS

Four pillars that make open source powerful



Transparency

Source code is fully visible to anyone. Users can inspect how software works, verify security, and understand exactly what the program does.

30M+ lines in Linux kernel - all public



Security

Global community rapidly patches vulnerabilities. With thousands of eyes reviewing code, security flaws get identified and fixed faster than proprietary software.

Avg fix time: 24-48 hours



Customization

Modify code for specific needs. Organizations can tailor software to their exact requirements without vendor restrictions or waiting for official updates.

800+ Linux distributions exist



Community Support

Massive global developer networks provide free support through forums, documentation, and direct code contributions. No vendor lock-in.

Linux: 20,000+ contributors

These features create a robust ecosystem where software evolves faster, stays more secure, and serves diverse needs

What is Proprietary Software

Closed-Source Software

Software where source code is kept secret by the vendor. Users cannot view, modify, or redistribute the code. Bound by strict End User License Agreements (EULAs).

"You don't own the software, you license it"

Key Characteristics

- **No Source Code Access** - Black box operation
- **License Restrictions** - Limited installations
- **Vendor Controlled** - Company decides features
- **Paid Licensing** - Per-user or subscription fees

Common Examples



Microsoft Windows

$199 for Pro version



Adobe Photoshop

$600/year subscription



Microsoft Office

$150/year subscription



macOS

Tied to Apple hardware



AutoCAD

$1,775/year subscription

Proprietary software generates $500B+ annually but limits user freedom

Problems with Proprietary Software

Why it creates real-world issues



High Cost

Expensive licenses and subscriptions create financial barriers. Enterprise software can cost thousands per user annually.

Windows 11 Pro: $199 | Adobe CC: $600/yr



Vendor Lock-in

Dependency on single company for updates, support, and compatibility. Difficult and expensive to switch to alternatives.

Migration costs: 3-5x license fees



Privacy Concerns

Data collection and telemetry without user consent. Companies monetize user data and behavior patterns.

Windows 10: 5,700 data points collected



Limited Customization

Cannot modify source code to meet specific needs. Users must wait for vendor to implement requested features.

Avg feature request: 2-3 years wait

These problems drive organizations toward FOSS alternatives that offer freedom, cost savings, and control

FOSS vs Proprietary Softwar

Head-to-head comparison across key dimensions

Feature	FOSS	Proprietary
Source Code Access	✓ Full Access	✕ Closed
Cost	✓ very low	✕ Lxpensive
Customization	✓ Unlimited	✕ Limited
Security	✓ High	✕ Variable
Privacy	✓ No Tracking	✕ Data Collection
Community Support	✓ Global	✕ Vendor Only
Vendor Lock-in	✓ None	✕ High Risk
License Freedom	✓ Full Freedom	✕ Restricted

FOSS Wins: 8/8

Cost Savings: 80-100%

Adoption Growing 15%/yr

What is an Operating System

System Software

An Operating System (OS) is system software that manages computer hardware, software resources, and provides common services for computer programs. It acts as an intermediary between users and hardware.

Core Functions

1. **Process Management**
   CPU scheduling, multitasking
2. **Memory Management**
   RAM allocation, virtual memory
3. **File System Management**
   Storage, organization, access
4. **Device Drivers**
   Hardware communication
5. **User Interface**
   GUI or command line
6. **Security & Access Control**
   User authentication, permissions

Popular Operating Systems



Windows
72% desktop

72%



macOS
15% desktop

15%



GNU/Linux
4% desktop

4%



Android
71% mobile

71%



iOS
28% mobile

28%

Linux dominates servers (96.4%) and supercomputers (100%) despite lower desktop share

What is GNU/Linux?

The perfect combination of GNU tools and Linux kernel



GNU Project

Started by Richard Stallman in 1983 to create a complete free Unix-like OS. Provides essential tools: GCC compiler, Bash shell, core utilities.

Free Software Foundation (1985)

Linux Kernel

Created by Linus Torvalds in 1991 as a hobby project. The kernel is the core that manages hardware resources and system processes.

Started as 65KB, now 30M+ lines

Development Timeline

1983 — **GNU Project**
Stallman starts

1985 — **FSF Founded**
Foundation

1991 — **Linux Kernel**
Torvalds

1992 — **First Distro**
MCC Linux

1993+ — **Distros Grow**
Debian, etc.

"GNU/Linux" is technically correct because Linux is just the kernel - GNU provides the essential OS tools

GNU/Linux vs Windows

Why Linux dominates in technical environments



Security

Linux: 2% malware infections vs Windows: 93%. Open-source code allows rapid vulnerability patching and transparent security auditing.

Linux wins: 47x fewer infections



Performance

Linux runs on 2GB RAM smoothly; Windows 11 needs 4-8GB. Linux boots in 20-30s vs Windows 30-45s.

2x more efficient resource usage



Customization

Modular architecture: choose kernel, desktop, package manager. Windows is monolithic with limited options.

800+ distros vs 1 Windows



Privacy

No telemetry or data collection. Windows collects 5,700+ data points. Linux respects user privacy by default.

Zero tracking vs extensive telemetry



Developer Friendly

Native containers, package managers, powerful terminal. Essential for modern development workflows.

Linux dominates: 96.4% servers, 100% supercomputers, 90% cloud

Why Learn Linux

Essential skills for modern engineering careers



Programming & Development

Native support for all languages

Powerful terminal, shell scripting, and development tools built-in. Supports Python, C++, Java, Go, Rust, and more natively.



AI & Machine Learning

Dominant OS for AI frameworks

TensorFlow, PyTorch, Jupyter all optimized for Linux. 90% of AI research runs on Linux systems.



Cybersecurity

Kali Linux & security tools

Kali Linux, penetration testing, network security analysis. Essential for ethical hacking and security careers.



DevOps

Docker, Kubernetes, CI/CD

Containerization and orchestration built for Linux. Native Docker/Kubernetes support makes deployment seamless.



Servers & Cloud Computing

90% of cloud infrastructure

AWS, Azure, Google Cloud all run Linux. Essential for cloud architects and system administrators. Linux skills are 2nd most in-demand in tech.

Linux skills: 2nd most in-demand | Avg salary: $80K-$200K+ | Job growth: 15% annually

Linux Outside Engineerin

Linux powers the modern world invisibly



Android

3 billion active devices run Linux-based Android, dominating 71% of the mobile market.

71% mobile market



Web Servers

96.4% of top 1M websites run Linux. Powers Instagram, Netflix, Amazon, and Google.

96.4% market share



Supercomputers

100% of TOP500 systems run Linux, handling climate modeling, nuclear simulations, and genomics.

100% dominance



ATMs & Banking

80% of stock exchanges and major banks use Linux for security and reliability.

80% financial sector



IoT & Embedded

Smart TVs, cars, routers, medical devices, and industrial controllers run embedded Linux.

Billions of devices



NASA & Space

ISS, Mars helicopter, SpaceX Falcon 9, and 32 space missions use Linux.

32 space missions

Linux is the invisible backbone of modern civilization - from your phone to outer space

Interesting Facts about Linux

Fascinating trivia about the world's most important OS



From Tiny to Massive

Started as 65KB hobby project in 1991, now 30M+ lines of code. Linus Torvalds' code is only 1% of the kernel today.



Tux the Penguin

Mascot named by James Hughes - "Tux" = Torvalds' Unix. Linus was bitten by a penguin at Canberra Zoo!



Steve Jobs' Offer

Steve Jobs offered Linus a job at Apple in 2000. Torvalds declined to continue working on Linux independently.



Space Missions

32 space missions use Linux, including NASA's Curiosity Rover, ISS computers, and SpaceX Dragon capsule.



Hollywood Loves Linux

Titanic was first film produced on Linux. Today 95% of Hollywood VFX and animation studios rely on Linux.



Asteroid Names

Asteroids named after Linux: 9793 Torvalds and 9885 Linux. Spacewatch recognized their impact!

800+ Distros

$5B Dev Investment

41,000 Person-Years

Popular Linux Distribution

Choose the right distro for your needs



Ubuntu

Beginner-friendly

Most popular Linux distro. Easy to use, great community support, LTS releases with 5-year support.

Best for: Beginners



Fedora

Developers

Cutting-edge technologies, latest software versions. Testing ground for Red Hat Enterprise Linux.

Best for: Developers



Debian

Stability

Rock-solid stability, 50,000+ packages. Foundation for Ubuntu and many other distros.

Best for: Servers



Kali Linux

Cybersecurity

600+ pre-installed security tools for penetration testing, ethical hacking, and forensics.

Best for: Security Pros



Arch Linux

Advanced

Do-it-yourself approach, rolling release, complete customization. Learn Linux from the ground up.

Best for: Power Users



Linux Mint

Windows-like

Familiar desktop experience for Windows migrants. Cinnamon desktop with multimedia support out-of-box.

Best for: Home Users

Each distro serves different needs - start with Ubuntu or Mint, then explore as you grow

Basic Linux Commands

Essential terminal commands to get started

Terminal Basics

ls	List files
cd	Change directory
pwd	Print working directory
mkdir	Make directory
rm	Remove files

File Operations

cp	Copy files
mv	Move/rename files
cat	Display file contents
chmod	Change permissions
grep	Search text patterns

System & Admin

sudo	Superuser privileges
apt/yum	Package manager
ps	List processes
top	System monitor
ssh	Remote login

Bash Shortcuts

Ctrl+C	Stop command
Ctrl+L	Clear screen
Tab	Auto-complete
↑/↓	Command history
Ctrl+R	Search history

Master these 15 commands and you'll be productive in any Linux environment

Career Opportunities with Linux

High-demand, high-paying career paths



Cloud Architect

Design cloud infrastructure

Design and manage cloud systems on AWS, Azure, GCP. Linux is foundational for all cloud platforms.

Salary: $150K-$200K+



DevOps Engineer

CI/CD, automation

Build deployment pipelines, manage containers, automate infrastructure. Linux-native tools.

Salary: $120K-$160K

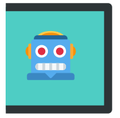


Cybersecurity Engineer

Security, pentesting

Penetration testing, security analysis, threat detection. Kali Linux and security tools.

Salary: $110K-$150K



AI/ML Engineer

Machine learning

Build AI models with TensorFlow, PyTorch on Linux. GPU computing and data pipelines.

Salary: $130K-$180K



Linux System Administrator

Server management

Manage servers, troubleshoot systems, ensure uptime. Foundation for all Linux careers. Certifications: RHCE, CompTIA Linux+, LFCS.

2nd Most In-Demand

15% Annual Growth

$80K-$200K+ Range

Conclusion

Linux represents freedom, collaboration, and innovation. It's not just an operating system - it's a philosophy that empowers users to control their technology.

🗽 Freedom
Use, study, modify, share

🤝 Collaboration
Global community

💡 Innovation
Rapid development

Essential skill for modern engineering: From AI to cybersecurity, cloud to DevOps, Linux opens doors to the most exciting and lucrative careers in technology.

Join the global open-source community. Start with Ubuntu or Linux Mint, master the terminal, and contribute to shaping the future of technology.



The future is open source. The future is Linux.

Q&A

Questions & Discussion

Thank you for your attention! I'm happy to answer any questions about FOSS, GNU/Linux, distributions, commands, or career opportunities.

🐧 Which distro to start?
Ubuntu or Linux Mint

💻 How to install Linux?
USB boot, virtual machine

🚀 Career paths?
Cloud, DevOps, Security

📚 Learning resources?
Linux Foundation, man pages

?

Let's discuss!



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SIGNAL



- **Free and Open Source Software**
- **Secure messaging and calling app**
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- **Protects user privacy and data**
- **Available on Android, iOS, Windows, Linux, and macOS**
- **No ads or user tracking**
- **Developed by the Signal Foundation**
- **Trusted worldwide for private communication**